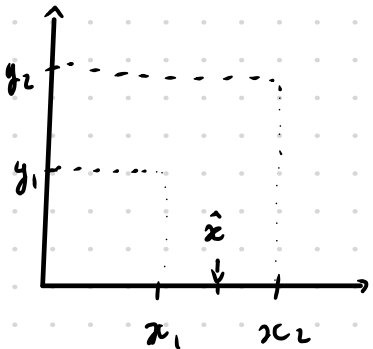


$$y = mx + c$$

$$= \left(\frac{y_2 - y_1}{x_2 - x_1} \right) x + c$$

$$y - y_1 = \left(\frac{y_2 - y_1}{x_2 - x_1} \right) (x - x_1)$$

$$\Rightarrow y = \left(\frac{y_2 - y_1}{x_2 - x_1} \right) (x - x_1) + y_1$$

$$= \frac{(y_2 - y_1)(x - x_1) + y_1(x_2 - x_1)}{x_2 - x_1}$$

$$= \frac{y_2(x - x_1) - y_1(x - x_1) + y_1(x_2 - x_1)}{x_2 - x_1}$$

$$= \frac{(x - x_1)y_2 + (x_2 - x)y_1}{x_2 - x_1}$$

$$\therefore y = \frac{(x - x_1)y_2 + (x_2 - x)y_1}{x_2 - x_1}$$

$$\hookrightarrow \hat{y} = \frac{(\hat{x} - x_1)y_2 + (x_2 - \hat{x})y_1}{x_2 - x_1}, \text{ let } h = x_2 - x_1$$

$$\text{Consider } \hat{y} = \sin(h): \quad \sin(1) \approx \frac{(1 - x_1)y_2 + (x_2 - 1)y_1}{x_2 - x_1}$$

$$= \frac{(1 - x_1)\sin(x_2) + (x_2 - 1)\sin(x_1)}{x_2 - x_1}$$

$$x_2 - x_1 = \hat{x}$$

$$\text{e.g. } x_1 \longleftrightarrow \hat{x} \longleftrightarrow x_2$$

$$x_1 \longleftrightarrow \hat{x} \longleftrightarrow x_2$$